

Week 11 — Achievers / HOTS Mixed Practice — Answer Key

Q1. A) 12 — If the number is x : $3x = x + 24 \rightarrow 2x = 24 \rightarrow x = 12$.

Q2. C) 25 — Three consecutive numbers summing to 72: let them be $n-1$, n , $n+1$; their sum is $3n = 72$, so $n = 24$, and the largest is 25.

Q3. A) ₹1,750 — Perimeter = $2 \times (20 + 15) = 70$ m; cost = $70 \times ₹25 = ₹1,750$.

Q4. B) ₹600 — After spending $2/5$, $3/5$ of the money remains, and that equals ₹360. So the total = $360 \times 5 \div 3 = ₹600$.

Q5. C) 8 — $10 \text{ L} = 10,000 \text{ mL}$; $10,000 \div 1,250 = 8$ jugs.

Q6. C) 13 — Let the younger sister's age be y ; the elder is $y + 4$. $y + (y+4) = 22 \rightarrow 2y = 18 \rightarrow y = 9$, so the elder is 13.

Q7. B) 250 kg — $3/10$ of 500 = 150 kg sold Monday; $1/5$ of 500 = 100 kg sold Tuesday; total sold = 250 kg; remaining = $500 - 250 = 250$ kg.

Q8. C) 16 — The gaps between terms increase by 1 each time (+1, +2, +3, +4, +5): 1, 2, 4, 7, 11, 16.

Q9. B) 88 — Number = (divisor $\times$ quotient) + remainder = $(5 \times 17) + 3 = 85 + 3 = 88$.

Q10. C) 100 — Roses = $1/4 \times 240 = 60$; marigolds = $1/3 \times 240 = 80$; lilies = $240 - 60 - 80 = 100$.

Q11. C) ₹300 — Amounts: 180, 220, 260, 300 (each ₹40 more than the last); check: $180+220+260+300 = 960$. ✓

Q12. B) 10 cm — Perimeter = $2 \times (\text{length} + \text{breadth}) = 48 \rightarrow \text{length} + \text{breadth} = 24 \rightarrow \text{breadth} = 24 - 14 = 10$ cm.

Q13. C) 20 — Total = people ahead + people behind + Rani herself = $8 + 11 + 1 = 20$.

Q14. B) 1,290 L — Combined rate = $250 + 180 = 430$ L/hour; in 3 hours = $430 \times 3 = 1,290$ L.

Q15. B) 38 — Let the number be $10a + b$, with $a + b = 11$. Reversing gives $10b + a$, and $(10b+a) - (10a+b) = 9(b-a) = 45 \rightarrow b - a = 5$. Solving with $a + b = 11$: $a = 3$, $b = 8$, so the number is 38. Check: reversed = 83, and $83 - 38 = 45$. ✓

Scoring: 12+/15 is a strong Achievers-section result — that's genuinely olympiad-level performance at this stage. Below that, go back to whichever Phase-1 week the missed questions came from before attempting Sample Paper 1.
